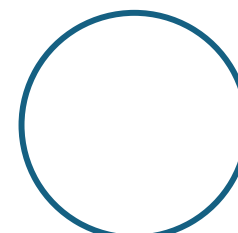
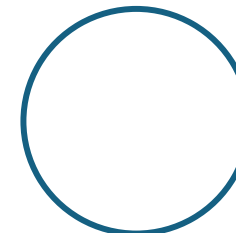




Benjamin Liu



Sergius Ching



Joyce Li

Client profile



Provider: NKE Private Wealth

Prestigious single-family office for the Chak family

Third-generation business dynasty

Ops across Japan and Greater China

Sectors: tech infrastructure, renewables, real estate

Family net worth > USD 2Bn

Client: The Chak Family

Patriarch has handed operations to G3 and is now focused on strategic asset allocation and legacy planning in a volatile market.

Client Mandate

Allocate USD 100 Mn in a product that compounds capital for future generations and captures Asia’s digital transformation

Goals



1 Capital for the next generations

Grow the USD 100mn ticket while keeping a defined capital barrier so the legacy sleeve is not an open-ended equity drawdown.



2 Digital transformation in Asia

Meaningful exposure to the compute – power–platform stack that sits next to the family’s own tech-infrastructure and renewable plants.



3 Bridge old Asia and new tech

Prefer underlyings that connect Japan and Greater China real-economy franchises with emerging technology

Risk tolerance: Moderate

Assumption

1

Family is open to structured notes / OTC, provided barriers and issuer credit are explicit.

2

Horizon of 5 years. Trade date 17 Sep 2026. Secondary liquidity is accepted as limited (issuer bid, not an exchange).

3

NKE / Natixis can design, price and book a bespoke note. Underlyings must be Bloomberg-searchable large caps; no OFAC names.

The family already owns Asian infrastructure, renewables and property: we add the AI hardware layer, not more of what they own

Our client should consider macro, geopolitical, and AI market factors

Geopolitics

US elections

- US midterms (Nov 2026) and the 2028 race keep trade policy uncertain
- US–China tech restrictions and tariffs hit cross-border families
- Capital rotates to supply-chain security and reshoring

Middle East, oil, inflation

- US , Iran conflict and Strait of Hormuz risk drive oil volatility
- Sticky core inflation keeps long-end yields high
- Rewards real pricing power; penalises cash-burning growth

AI pivots



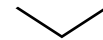
Most of the money is not in models

Worldwide AI spend is about USD 2.7tn in 2026. Infrastructure takes roughly half.



Building ahead of demand

Many projects are still pilots. The cycle only pays off if AI creates new real-economy revenue (factories, grids, science)



Client already owns plants and infrastructure in Asia sits closer to the scarce part of the market

Why hardware products ?



Software

- Cycle faced a valuation rationalization
- High-multiple AI software de-rates on higher discount rates



Hardware

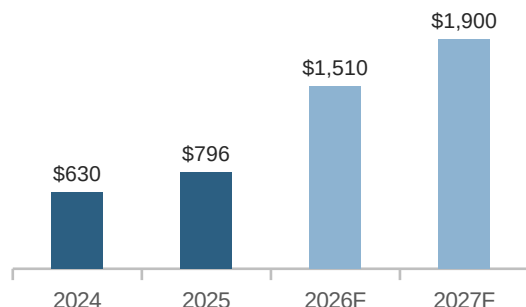
- AI performance is no longer driven by coding apps, but by hardware, localized grid/power infrastructure etc.
- Capex shifts decisively to chips, HBM memory, equipment, servers

The client should own Asia's AI hardware, with protection for the legacy and income while we wait

Investment thesis 1: Asia controls AI's hardware bottlenecks

1a Demand is structural, and it runs through Asia

Global chip market, US\$bn (WSTS)



Demand is structural

Hyperscaler capex ~US\$725bn in 2026 (+77%); the chip market nearly doubles to ~US\$1.5trn.

The moat runs through Asia

8 of the world's 10 largest companies depend on TSMC (J.P. Morgan).

...but the cycle is maturing

DRAM price gains slowing: +48% in Q2, ~+13% expected in Q4 2026.

~70%

Taiwan makes the chips: TSMC's share of global foundry revenue; TSMC is >40% of TAIEX.

→ Taiwan is the single largest AI chokepoint

83%

Korea makes the memory: SK hynix + Samsung share of HBM, the memory every AI accelerator needs.

→ No AI accelerator ships without Korean memory

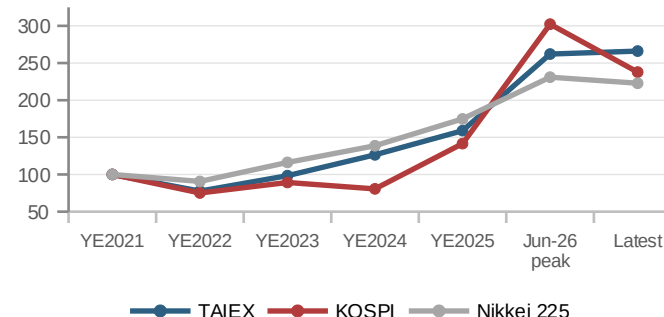
~55%

Japan makes the tools: Advantest's share of chip test equipment; with Tokyo Electron ~ 20% of the Nikkei.

→ Every new fab needs Japanese tools

1b A strong theme on a violent path

Index level, end-2021 = 100



Growth is structural...

TAIEX and KOSPI are 2–3× their end-2022 levels on AI demand.

...but the path is violent

Down 10–25% in 2022; records in June 2026, then sharp falls within weeks.

–25%

Round trips happen fast: KOSPI went from 9,000+ in June to 6,789 on 28 Aug 2026, in about ten weeks.

→ Entry timing is hard; drawdowns are deep and quick

>40%

Returns hinge on a few names: TSMC >40% of TAIEX; Samsung + SK hynix ~48% of KOSPI; ~92% of advanced chips from Taiwan.

→ Single-name risk is high

38.6×

Valuations price in a lot: Nikkei CAPE 38.6×; Taiwan among the most expensive markets; DRAM gains slowing.

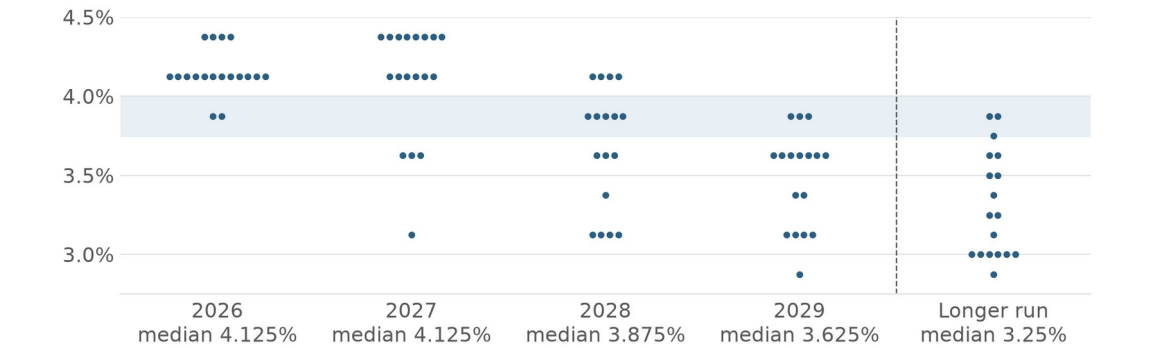
→ Upside from here is less certain than the theme

Whichever AI company wins, it needs Asia's chips, memory and tools, but the path to owning them is violent.

Investment thesis 2: Rates stay high and volatility is at records

The backdrop for the next five years: the Fed holds rates high, bonds carry rate risk, and Asian equities swing harder than ever

2a The Fed holds rates high for longer



FOMC dot plot, 16 Sep 2026 (shaded: today's 3.75–4.00%). Dots read from the Fed's chart.

4.125%

FOMC median to end-2027, cuts deferred to 2028: 12 of 18 dots at 4.125% for 2026; 4.375% seen by 4 for 2026 and 8 for 2027.
→ Rate risk is structural, not a one-off

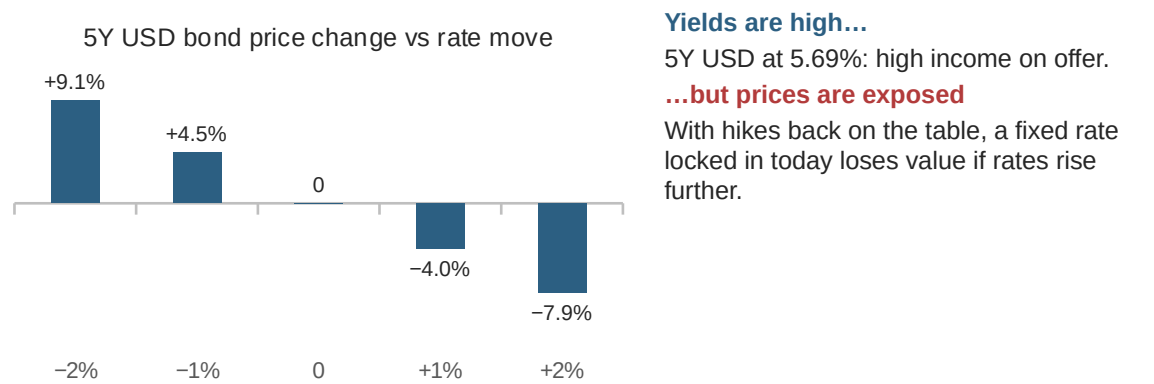
+0.25%

The Fed is hiking again: 16 Sep 2026: to 3.75–4.00%, the first hike since 2023.
→ The easing cycle is over for now

~87%

Markets price more: probability of at least one further hike priced by fed funds futures.
→ Risk to rates is still to the upside

2b Bonds carry rate risk, equities swing hard



-4.0%

Long bonds carry rate risk: a 5-year bond loses ~4.0% per 1% rise in rates.
→ Duration is a risk, not a hedge, in this cycle

91.2

Volatility is at records: VKOSPI closed at an all-time high of 91.2 (9 Jun 2026), above the 2008 peak.
→ Option-implied risk in Asian equities is extreme

-8.95%

Single days move markets: KOSPI's one-day fall on 13 Jul 2026; J.P. Morgan expects a 10–15% correction in 2026.
→ Buy-and-hold must sit through deep drawdowns

Rates stay high for longer and Asian equity volatility is at records: neither bonds nor buy-and-hold are a comfortable place to sit.

Our Product : 12.75% p.a. 5-Year Asia Mixed Technology Autocallable Note

Why this design: a double-digit coupon, a 70% barrier at maturity, and capital usually back early

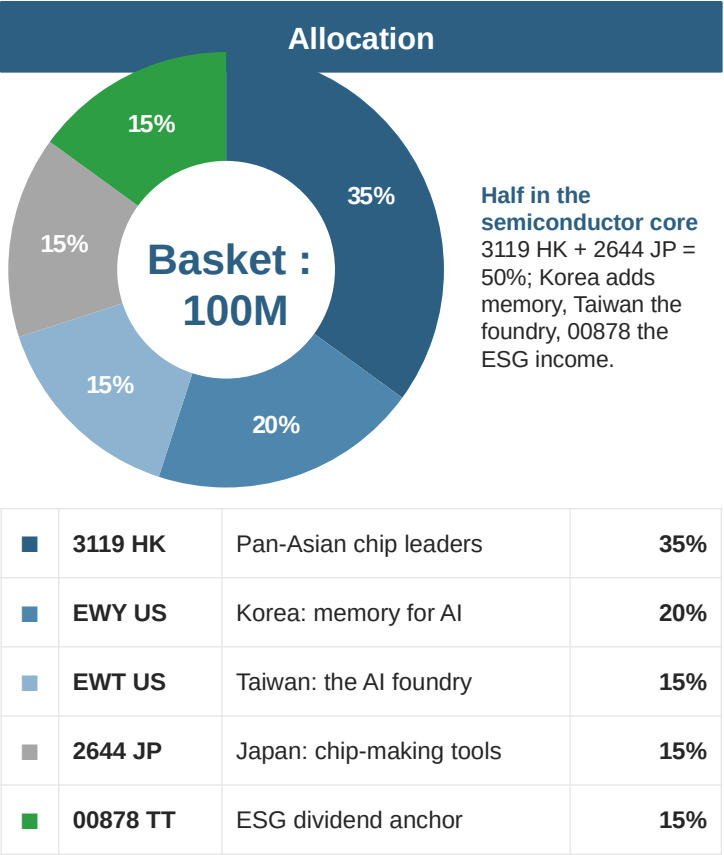
Issuer	Natixis SA (BPCE Group)
Notional / currency	USD 100,000,000
Trade / maturity	17 Sep 2026 / 17 Sep 2031 (5Y)
Underlying	Weighted basket of 5 Asian ETFs
Observation	Yearly, 5 dates; autocall from Year 1
Coupon	12.75% p.a., paid at call: $100\% + 12.75\% \times \text{years elapsed}$
Autocall	Called in year k if basket $\geq 105\% - 2.5\% \times k$ (102.5%, 100%, 97.5%, 95%, 92.5%)
Maturity barrier	70%, observed at Year 5 only: if never called and the basket ends below 70%, repaid at the basket level; otherwise 100%
Issue price / fair value	100% / 98.4% (fair coupon 14.09%)

ETF (Bloomberg)	Wt	Role in the basket
Asia Semiconductor (3119 HK)	35%	Pan-Asian chip leaders (40 names; TSMC, Samsung, SK Hynix the largest): the core AI-hardware exposure
iShares MSCI South Korea (EWY)	20%	Korea large caps (Samsung + SK hynix ~44%): high-bandwidth memory for AI
iShares MSCI Taiwan (EWT)	15%	Taiwan large caps (TSMC, MediaTek, Delta): AI foundry and server chain in the family's Greater China region
Japan Semiconductor (2644 JP)	15%	Japanese semiconductor stocks (FactSet Japan Semiconductor Index, e.g. Tokyo Electron, Advantest): the family's home market
Taiwan ESG High Dividend (00878 TT)	15%	ESG-screened Taiwan high-dividend leaders (Quanta, UMC, MediaTek): ESG sleeve and lower-volatility anchor

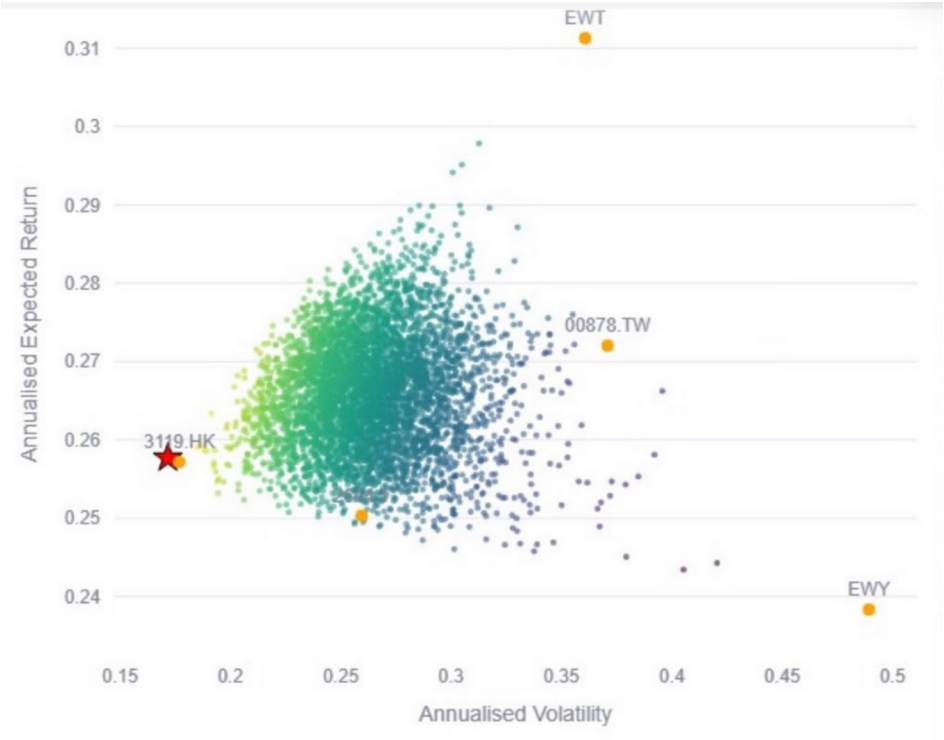
- **Structure:** Autocallable Note.
- **Asset Allocation:** Applied Modern Portfolio Theory (MPT) and Relative Value (RV) analysis to optimize weights for a 5-ETF underlying basket (*details on next slide*).
- **Pricing & Valuation:** Combined Implied Volatility (IV) with Monte Carlo simulations to derive the fair coupon price (*methodology in Appendix E & F*).

Strategy: our five-ETF basket sits at the low-risk edge of every possible mix

USD 100mn across five Bloomberg-listed ETFs; our weights (red star) deliver ~26% expected return for only ~17% Realized Volatility



Modern Portfolio Theory : Risk vs return: single ETFs and random mixes



How to read it

Orange dots: each ETF on its own (2644 is the dot near 0.26, 0.25).

Red star: our basket (35 / 20 / 15 / 15 / 15).

Small dots: thousands of random weightings of the five ETFs; lighter = more return per unit of risk.

Axes: annualised volatility (risk) and expected return.

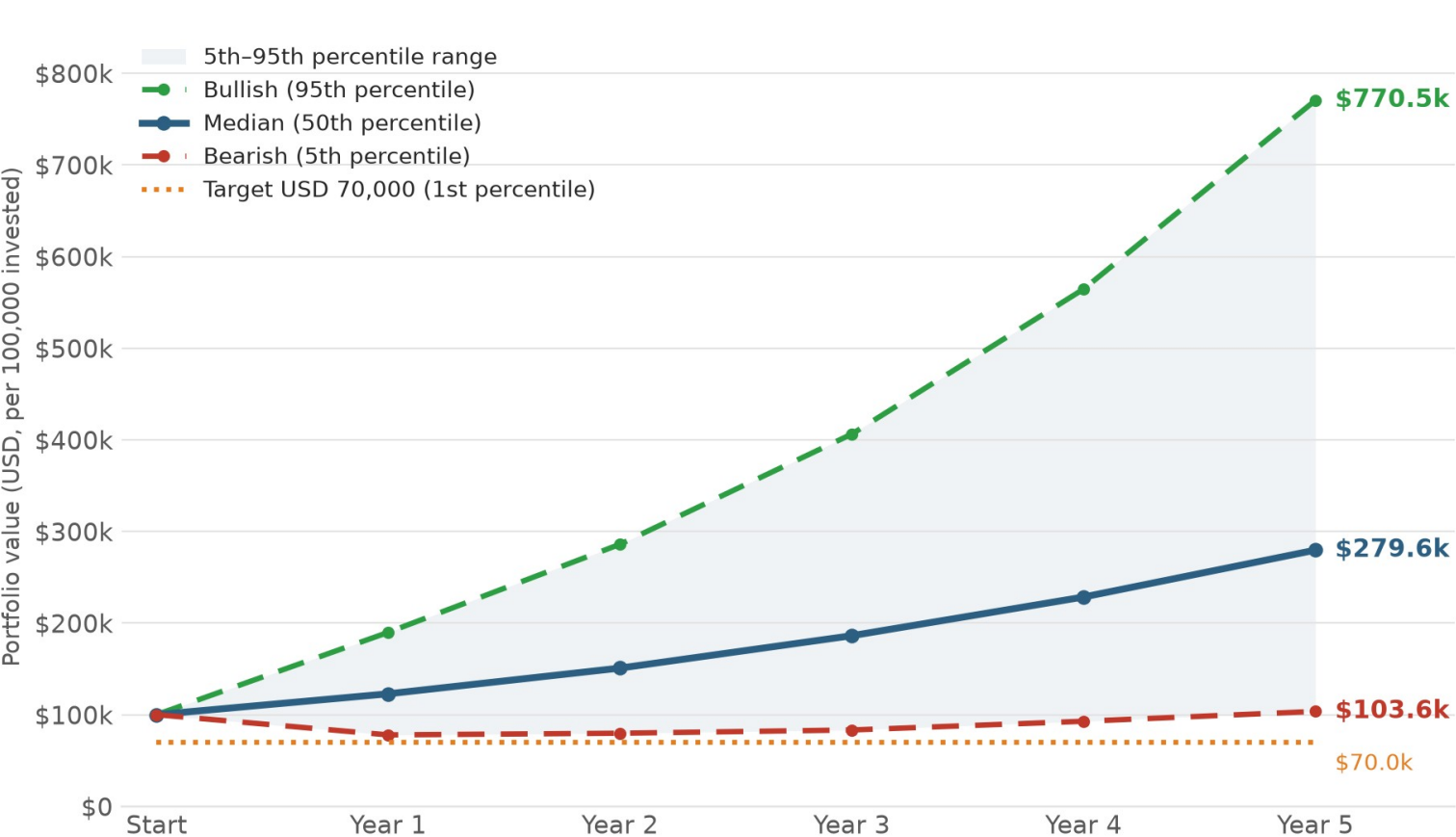
Takeaways

- Our basket (red star): ~26% return at ~17% volatility, the lowest risk of all the mixes shown.
- Most random mixes need 22–32% volatility for the same 25–28% return: our weights get it with less risk.
- Less risk than any single ETF, about a third of EWY's (~49%), while giving up only a few points versus EWT's ~31%.
- Why: 3119 HK (35%) anchors the basket; the other four move differently enough to pull risk down.

Our basket (red star): ~26% return for ~17% volatility, less risk than any single ETF or random mix

Monte Carlo: the median portfolio grows 2.8× in five years

5-year simulation of the five-ETF portfolio, per USD 100,000 invested (×1,000 for the USD 100mn mandate)



\$279.6k

Median at Year 5: 2.8× the start, +22.8% a year

\$103.6k

Bearish (5th percentile): still above the start, +0.7% a year

\$770.5k

Bullish (95th percentile): 7.7× the start

99.0%

Chance of ending at or above the USD 70,000 target

Monte Carlo methodology

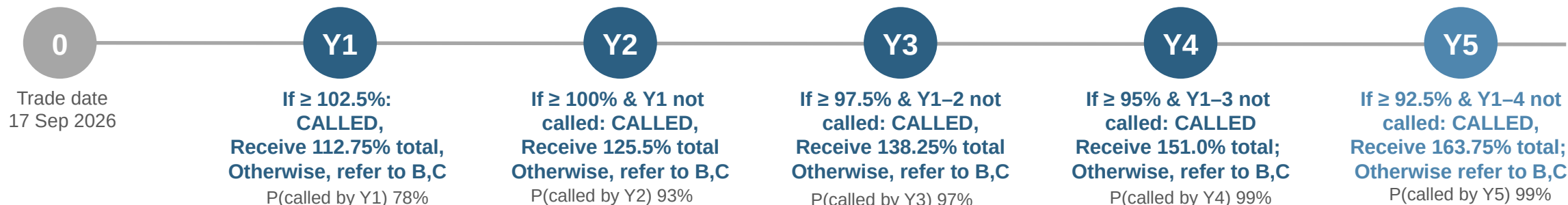
- **Historical resampling:** past 5 years of monthly returns, drawn at random with replacement.
- **Path generation:** 10,000 unique 5-year paths, each built from 60 monthly returns.
- **Payoff valuation:** each path runs the note's exact autocall, coupon and 70% barrier logic.
- **Risk & return metrics:** aggregated into call probabilities, barrier breaches and the payout distribution.

Even the bearish path ends above the start (\$103.6k); the USD 70,000 target is the 1st percentile

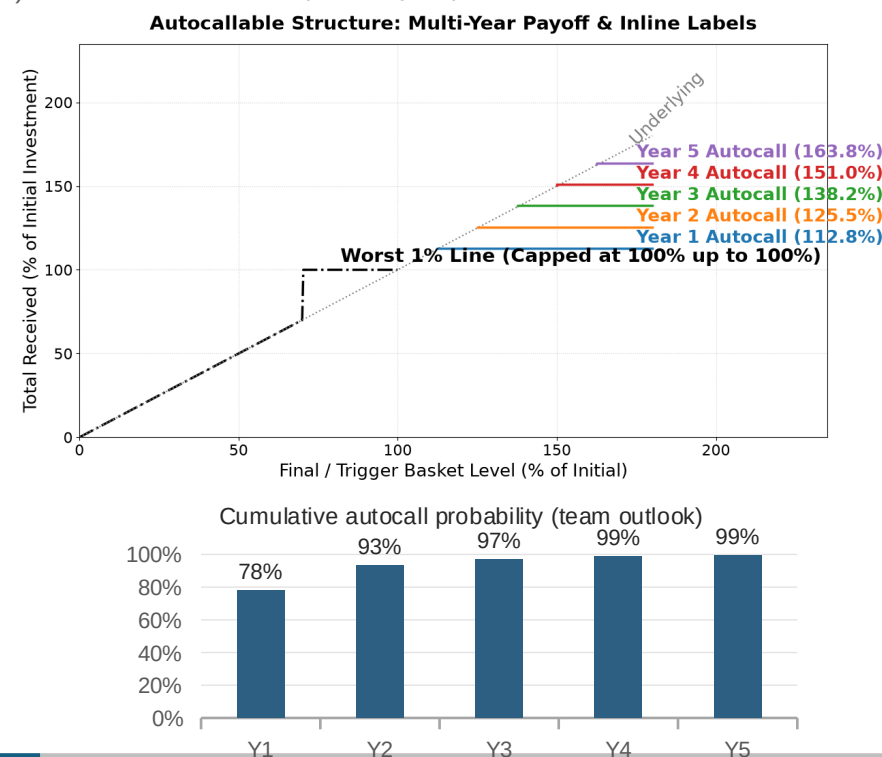
How it pays: one question every year, three outcomes

Is the basket at or above that year's call level? (checked once a year: $105\% - 2.5\% \times \text{year}$)

Call levels step down from 102.5% (Y1) to 92.5% (Y5). Called in year k: $100\% + 12.75\% \times k$. No coupon before the call.



- A Autocalled**
Basket $\geq$ that year's call level: $100\% + 12.75\% \times \text{years}$. The note ends.
99.3% called within 5 years
- B Not called, basket $\geq 70\%$ at Y5**
Years 1–4: continue holding
Year 5: repaid at 100% of notional
0.5% not called but $\geq 70\%$ at Y5
- C Not called, basket $< 70\%$**
Repaid at the basket level at Y5: the family bears the full fall.
0.2% end below 70% at Y5



Expected value at exit (team outlook): ~114.2% of notional

Risks for the family, and how Natixis hedges its side

Risk	What could happen	Mitigant
Capital loss	Never called and basket < 70% at Y5: repaid at the basket level (0.2% on the team outlook; ~14% under pricing assumptions, average loss ~50%)	Diversified basket; 70% barrier at maturity only; ~5% of family wealth
Capped upside	Team median is 2.8× in 5 years; the note pays 12.75% p.a. and is usually called by Y1	Coupons lock in gains if the rally stalls; rest of family wealth keeps direct equity exposure
ETF structure	00878 pays ~7.6% a year out, lowering its price path; 3119 trading is thin; ETF tracking error	00878 held at 15%; verify volumes; Natixis can reference total-return versions
Issuer credit	Coupons and repayment depend on Natixis / BPCE	BPCE senior rating; optional collateralised wrapper
Liquidity / MTM	Sold before maturity, the price can be below 100%	Natixis daily indicative price; hold to call or maturity

Pricing sensitivity (fair coupon, annual model):

- Base case: $C_{\text{fair}} = 14.09\%$
- Volatility shift: $\sigma \pm 3\text{pts} \Rightarrow C_{\text{fair}} \in [12.6\%, 15.6\%]$, $\partial C / \partial \sigma > 0$
- Correlation shift: $\rho -0.15 / +0.14 \Rightarrow C_{\text{fair}} \in [13.0\%, 15.0\%]$, $\partial C / \partial \rho > 0$
- Equity drift: $\Delta\mu = -1\% \text{ to } -2\% \Rightarrow C_{\text{fair}} = 15.3\% - 16.7\%$
- The 12.75% offer stays below fair value in every case except vol -3pts (12.6%).

Coupon digitals

Natixis owes digital coupons on the step-down call levels (102.5% → 92.5%): replicated with tight basket call spreads, delta-hedged with ETF units, constituent stocks and index futures as proxies; listed options on EWY/EWT

Vol & correlation

Coupon value moves with basket vol and correlation (fair 12.6%–15.6% for $\sigma \pm 3\text{pts}$): managed with listed index options and correlation trades

Quanto FX

EWT and EWY already trade in USD; 3119 (HKD, pegged), 2644 (JPY) and 00878 (TWD) need FX forwards and a quanto adjustment

Short put & funding

Natixis is long the client's put: struck at 100%, live only if never called and the basket ends below 70% at Y5; hedged with basket put spreads and delta. Principal funded at 5.69%

Appendix A: pricing assumptions and methodology

Inputs from Appendix F (implied vols, correlations); the strategy app (slides 7–8) uses historical data and is not used for pricing

ETF	Weight	Vol Y1–Y5	Div	3119 HK	EWY US	EWT US	2644 JP	00878 TT
3119 HK	35%	30–38%	0%	1.00	0.53	0.74	0.86	0.56
EWY US	20%	42–45%	0%	0.53	1.00	0.79	0.69	0.44
EWT US	15%	32–36%	0%	0.74	0.79	1.00	0.61	0.69
2644 JP	15%	23–26%	0%	0.86	0.69	0.61	1.00	0.52
00878 TT	15%	7–12%	0%	0.56	0.44	0.69	0.52	1.00

Model. Correlated GBM for each ETF, 1,000,000 paths, annual steps, implied-vol term structure through interval variances (App. F). Risk-neutral drift = USD funding; dividends set to zero. Quanto adjustment ignored (assumed hedged by the desk).

Discounting. Case USD funding grid; 5Y discount factor 0.758.

Coupon solve. Fair coupon makes the note worth exactly 100% (App. E): 14.09%. We offer 12.75%; the 1.34% a year difference is Natixis' hedging cost and margin.

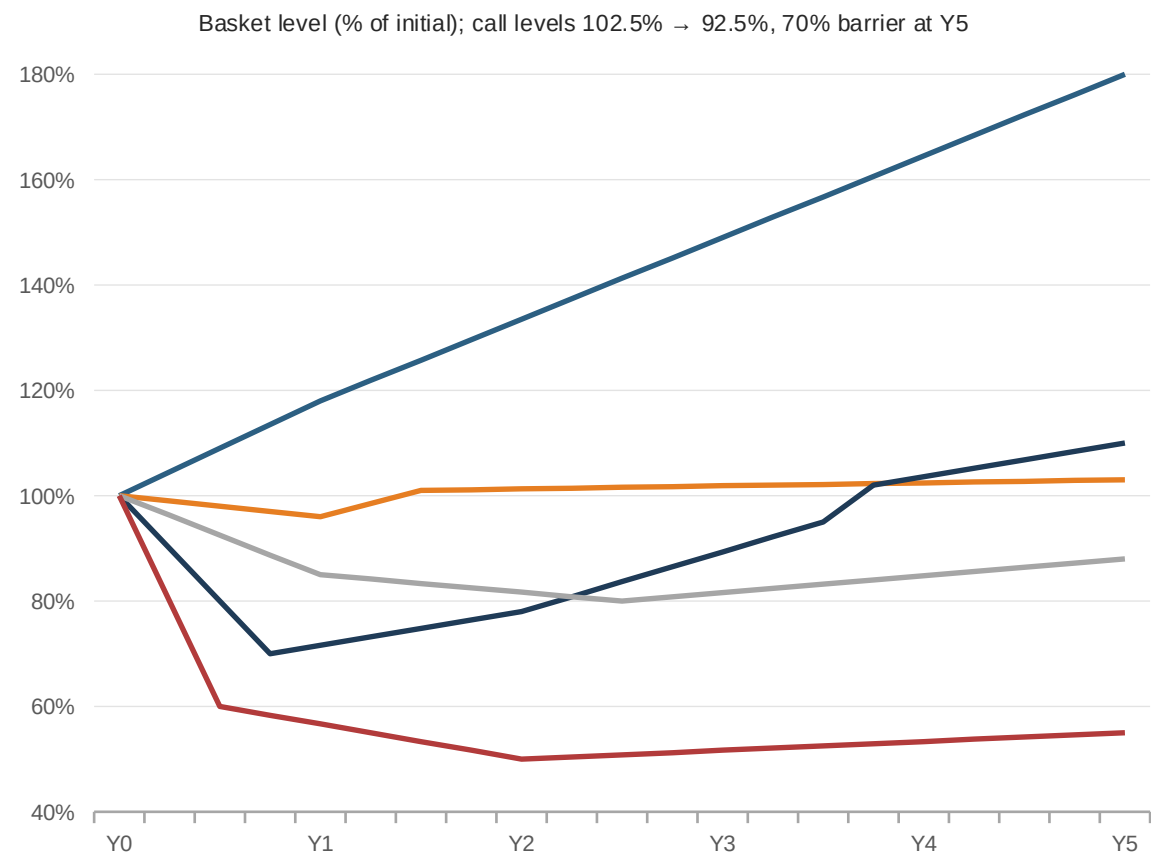
Historical back-test (to run). Monthly Bloomberg history since 2021 for all five ETFs. Launch a hypothetical note every month, apply the same call levels and barrier, and record coupons, life and IRR by launch year.

Variant of our note	Fair cpn
70% barrier at Y5 (base)	14.1%
Barrier 65%	13.8%
Barrier 75%	14.3%
No barrier	14.7%
Equity drift –1%	15.3%
Equity drift –2%	16.7%
Vol +3pts / –3pts	15.6% / 12.6%

Variants re-run on the annual model (400k paths) and shown as shifts from the 14.09% base. Code: `pricing/scenario2_annual_checks.py`

Appendix B: scenario analysis, five paths for Asian markets

Hypothetical basket paths, checked yearly against the call levels (102.5% → 92.5%) and the 70% barrier at Y5



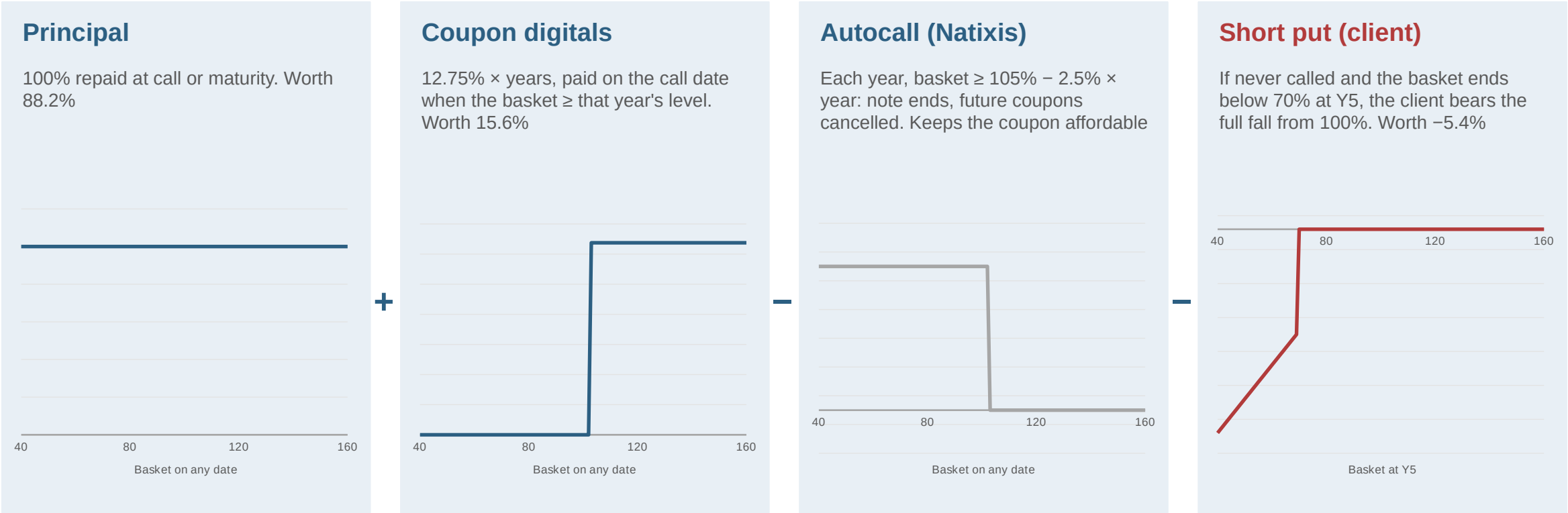
Scenario	Outcome	Coupons	USD mn	IRR	Basket
AI capex boom	Called Y1 (118% ≥ 102.5%)	12.75%	112.75	12.75%	–
Range-bound Asia	Called Y2 (101% ≥ 100%)	25.5%	125.5	12.0%	–
2022-style tech drawdown, slow recovery	Called Y4 (104% ≥ 95%)	51.0%	151.0	10.9%	–
Lost decade (sideways below par)	Not called; 88% ≥ 70%; 100% back	0.0%	100.0	0.0%	–12%
Geopolitical shock, no recovery	Not called; 55% < 70%	0.0%	55.0	–11.3%	–45%

Reading: the note earns 10.9%–12.75% a year whenever the basket reaches that year's call level, even after a 30% drawdown (called in Y4). In the lost decade the basket ends at 88%, above the 70% barrier, so the family gets 100% back; only a lasting fall below 70% passes the loss through (55%).

Paths are illustrative and not forecasts. 'Basket' = direct basket return at Y5 when the note is not called. Historical back-test to run on Bloomberg history (appendix).

Appendix C: how Natixis replicates the note

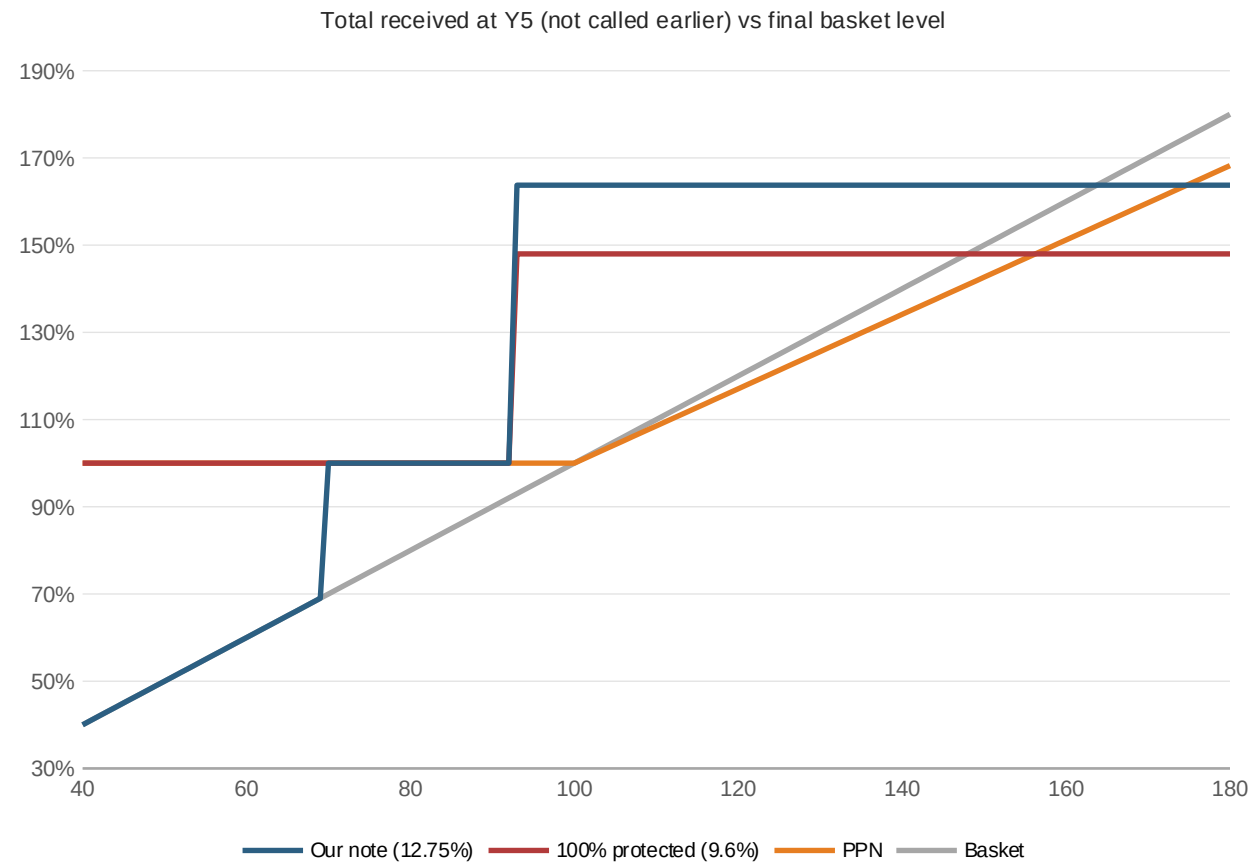
Replication used by the Natixis desk to price and hedge (values in % of notional, at trade date)



Fair value 98.4% = principal 88.2% + coupons 15.6% – short put 5.4%. Issued at 100%, leaving 1.6% (≈1.34% a year) for hedging and margin. The short put funds the double-digit coupon.

Appendix D: alternatives for the investment committee

Same basket and call levels: full protection, no barrier, and a pure participation note



Option	Return	Loss risk
Step-down autocall (ours)	12.75% p.a. at call	14% pricing; 0.2% outlook
100% protected autocallable	~9.6% fair	0%
No barrier	~14.7% fair	~19% pricing
Principal-protected note	85% of gain at Y5	0%

Trade-off: full protection pays only ~9.6%. Removing the barrier adds ~0.6% of coupon but loses capital in ~19% of paths instead of ~14% (pricing measure). The PPN pays nothing unless the basket ends above 100% after 5 years. Our note pays 12.75%, with money usually back within ~1 year on the team outlook, and protects the family down to a 30% fall at maturity.

Annual Monte Carlo on App. F inputs (400k paths); fair coupons shown as shifts from the 14.09% base. Payoffs at Y5 for notes not called earlier. Inputs to refresh with Bloomberg data as of the 17 Sep 2026 trade date.

Appendix E: how we priced the note, and why the fair coupon is 14.09%

Monte Carlo simulation of the five-ETF basket (1,000,000 paths); the fair coupon makes the note worth exactly its issue price

A	Term structure & discount factors Annual rates are converted to continuous rates; each cash flow is discounted at its own tenor	$r_j = \ln(1 + R_j) \quad (1)$ $D_j = \exp(-r_j \cdot t_j) \quad (2)$
B	Asset paths (correlated geometric Brownian motion) Each ETF moves one year at a time with its own interval variance ΔV ; shocks are correlated through the Cholesky factor L of ρ	$X_{i,j} = X_{i,j-1} \cdot e^{\left((r_j - q_{i,j} - \frac{1}{2}\Delta V_{i,j}) + \sqrt{\Delta V_{i,j}} Z_{i,j}\right)} \quad (3)$ $Z = \varepsilon \cdot L^T, \quad \rho = L \cdot L^T \quad (4)$
C	Basket & autocall trigger The basket is the weighted sum of normalised ETF prices; the note ends at the first year it reaches that year's barrier	$B_j = \sum_{i=1..5} w_i X_{i,j} \quad (5)$ $\text{call at first } k \text{ with } B_k \geq H_k \quad (6)$
D	Maturity payout (never called) A 70% barrier observed only at Year 5	$R_5 = 1.0 \text{ if } B_5 \geq 0.70; \quad R_5 = B_5 \text{ if } B_5 < 0.70 \quad (7)$
E	Fair coupon a = discounted principal repaid, b = discounted years of coupon (Case A: no coupon on never-called paths)	$1.0 = E[a + y \cdot b] \quad (8)$ $y = \frac{(1.0 - E[a])}{E[b]} \quad (9)$

Product priced		
Basket weights	3119.HK 35% · EWY 20% · EWT 15% · 2644.T 15% · 00878.TW 15%	
Autocall barrier H	105% – 2.5% × k: Y1 102.5% · Y2 100% · Y3 97.5% · Y4 95% · Y5 92.5%	
Called in year k	100 × (1 + k × y)	
Never called	R ₅ as in (7)	
Simulation	1,000,000 paths, annual steps	
Pricing results (Case A)		
Fair coupon		14.092219%
Standard error		2.236 bp
Approx. 95% CI	[14.048393%, 14.136045%]	
14.092219% fair = 12.75% coupon to the client + 1.342219% kept by Natixis		

We keep 1.342219% a year and give the client a 12.75% coupon

Source: team write-up 'Derivation of Fair Coupon Rate for Autocallable Structured Product' (8 Oct 2026), sections 2–3. Excludes issuer credit, fees and hedging costs.

Appendix F: pricing inputs, grids and matrices

Every input used in the Monte Carlo simulation (team write-up, section 4)

4.1 Term structure of interest rates			
Year (t)	Risk-free rate (R)	Continuous rate $r = \ln(1 + R)$	Discount factor (D)
1	4.78%	4.669%	0.9544
2	5.30%	5.164%	0.9019
3	5.52%	5.373%	0.8511
4 (interpolated)	5.605%	5.454%	0.8040
5	5.69%	5.534%	0.7583

4.2 Correlation matrix (ρ)					
Asset	3119.HK	EWY	EWT	2644.T	00878.TW
3119.HK	1.000	0.534	0.741	0.863	0.562
EWY	0.534	1.000	0.787	0.691	0.438
EWT	0.741	0.787	1.000	0.614	0.689
2644.T	0.863	0.691	0.614	1.000	0.517
00878.TW	0.562	0.438	0.689	0.517	1.000

4.3 Implied volatility grid (σ), annualised					
Asset / Year	Year 1	Year 2	Year 3	Year 4	Year 5
3119.HK	30.45%	32.34%	33.50%	36.70%	37.50%
EWY	41.50%	42.70%	44.50%	44.50%	44.50%
EWT	32.40%	35.80%	36.30%	36.30%	36.30%
2644.T	23.00%	24.90%	25.60%	25.60%	25.90%
00878.TW	7.00%	9.00%	12.00%	12.00%	12.00%

4.4 Interval variances (ΔV), drive each year's diffusion step					
Asset / Interval	Year 1	Year 2	Year 3	Year 4	Year 5
3119.HK	0.0927	0.1165	0.1275	0.2021	0.1644
EWY	0.1722	0.1924	0.2294	0.1980	0.1980
EWT	0.1050	0.1514	0.1390	0.1318	0.1318
2644.T	0.0529	0.0711	0.0726	0.0655	0.0733
00878.TW	0.0049	0.0113	0.0270	0.0144	0.0144

Basket weights 3119.HK 35% · EWY 20% · EWT 15% · 2644.T 15% · 00878.TW 15% | Call levels 105% – 2.5% × year (102.5% → 92.5%) | Maturity barrier 70%

Source: team write-up, section 4. Rates: case USD funding grid (Year 4 interpolated); discount factors $D = \exp(-r \cdot t)$ and $\Delta V = t \cdot \sigma(t)^2 - (t-1) \cdot \sigma(t-1)^2$ recomputed from the tables. Correlation matrix checked positive definite.

Appendix G: References

† = secondary source, value read from a chart, or source still to be added: verify before submission. Slide numbers in brackets. Full links: [deck/REFERENCES.md](#)

Case and client

[1] Natixis CIB, Investment Strategy Challenge 2026, Scenario 2 brief: NKE Private Wealth / Chak family mandate, USD funding grid, product rules (2, 5, 10–16)

AI demand and the Asian hardware chain

[2] WSTS, Spring 2026 semiconductor market forecast (May 2026) and 2025 final release (4)

[3] Hyperscaler 2026 capex guidance (Amazon, Microsoft, Alphabet, Meta), compiled by AI Weekly and Yahoo Finance (4)

[4] Counterpoint Research, foundry revenue share (TSMC ~70%), via XTB and Motley Fool (4)

[5] Counterpoint Research, Global DRAM and HBM market share (SK hynix 50%, Samsung 33%, Q2 2026) (4)

[6] TrendForce, DRAM contract price outlook Q2–Q4 2026 (via iConnect007) (4)

[7] Trading Economics and Focus Taiwan, TSMC >40% of TAIEX market value (2026) (4)

[8] DataM Intelligence / Envisioning, Advantest ~55% of chip test equipment† (4)

[9] Nikkei Inc., Advantest weight in the Nikkei 225 and index archive (4)

[10] Worldwide AI spending ~USD 2.7trn in 2026 (slide 3): source not recorded by the team† (3)

Markets, valuation and volatility

[11] Cembalest, M., Eye on the Market 2026 Outlook: "Smothering Heights", J.P. Morgan Asset & Wealth Management, 1 Jan 2026 (pp. 1, 33–34) (4, 5)

[12] TWSE Market Highlights 2021–2025; Trading Economics (TAIEX 2026 record) (4)

[13] Korea Exchange year-end KOSPI levels 2021–2025 (via Korea Times, Invezz)† (4)

[14] BIT Research, "The KOSPI supercycle" (KOSPI 9,000+, Samsung + SK hynix ~48%)† (4)

[15] FN News, KOSPI close 6,789 on 28 Aug 2026 (4)

[16] Asiae, KOSPI –8.95% on 13 Jul 2026 (5)

[17] Nikkei Inc. index summary; Nippon.com / Jiji (Nikkei 225 levels) (4)

[18] Sibilis Research, Nikkei 225 CAPE (38.6×) and KOSPI P/E (4)

[19] SBS and Herald Corp, VKOSPI record close 91.23 (9 Jun 2026) (5)

Rates

[20] Board of Governors of the Federal Reserve System, Summary of Economic Projections, 16 Sep 2026 (dot plot; counts read from chart)† (5)

[21] Advisor Perspectives and Admiral Markets, FOMC decision of 16 Sep 2026 (+0.25% to 3.75–4.00%)† (5)

[22] Fed funds futures, ~87% probability of a further hike: source not recorded by the team† (5)

Underlying ETFs and data

[23] Global X ETFs (Hong Kong), Global X Asia Semiconductor ETF (3119 HK), product factsheet† (6, 7)

[24] BlackRock iShares, MSCI South Korea ETF (EWY) and MSCI Taiwan ETF (EWT), factsheets and holdings† (6, 7)

[25] Global X Japan, Global X Japan Semiconductor ETF (2644 JP), factsheet† (6, 7)

[26] Cathay SITE, Cathay Taiwan ESG Sustainability High Dividend ETF (00878 TT), factsheet† (6, 7, 10)

[27] Bloomberg L.P., PX_LAST daily prices for the five ETFs (team file data/Stock.xlsx) (7, 8, 9)

Methodology

[28] Markowitz, H. (1952), "Portfolio Selection", Journal of Finance 7(1), 77–91 (7)

[29] Efron, B. (1979), "Bootstrap Methods: Another Look at the Jackknife", Annals of Statistics 7(1), 1–26 (8, 9)

[30] Black, F. and Scholes, M. (1973), "The Pricing of Options and Corporate Liabilities", Journal of Political Economy 81(3), 637–654 (11, 15)

[31] Glasserman, P. (2004), Monte Carlo Methods in Financial Engineering, Springer (11, 15, 16)

[32] Hull, J. C., Options, Futures, and Other Derivatives, Pearson (10, 13)

[33] Bouzoubaa, M. and Osseiran, A. (2010), Exotic Options and Hybrids: A Guide to Structuring, Pricing and Trading, Wiley (6, 13, 14)

[34] Team write-up, "Derivation of Fair Coupon Rate for Autocallable Structured Product" (8 Oct 2026) and code pricing/stochastic_cal_for_natixis.py, pricing/scenario2_annual_checks.py (10–16)

Summary: Asia's AI hardware, held through an autocall that pays while markets swing

From the family's mandate to the note: what we recommend, why now, and what the family can expect

1 The client	2 The thesis	3 The product	4 The outcome
USD 100mn, 5 years, moderate risk Chak family office (NKE Private Wealth); ~5% of family wealth.	Asia controls AI's hardware ~70% of foundry (Taiwan), 83% of HBM (Korea), ~55% of chip test (Japan).	12.75% a year, 5Y autocallable note Natixis-issued, USD 100mn, five Asian ETFs: 3119 HK 35%, EWY 20%, EWT 15%, 2644 JP 15%, 00878 TT 15%.	Capital usually back early 78% called in year 1 (112.75%); 99.3% called within five years.
Two goals Compound capital for the next generation, and own Asia's digital transformation.	But the path is violent KOSPI -25% in ten weeks; VKOSPI at a record 91.2.	Step-down call, yearly Called if the basket $\geq 102.5\%$ in Y1, falling 2.5% a year to 92.5% in Y5; pays 100% + 12.75% $\times$ years.	Loss is rare 0.2% of paths end below 70%, where the family takes the basket's fall.
One constraint A defined capital barrier, not an open-ended equity drawdown.	And rates stay high FOMC median 4.125% to end-2027; long bonds lose ~4% per 1% rise.	70% barrier at maturity only If never called, 100% back unless the basket ends below 70%.	Priced with room Fair coupon 14.09%: Natixis keeps 1.34% a year for hedging and margin.

Why it fits the family

12.75%	Growth: a double-digit coupon from the AI-hardware theme, without having to time entry	70%	Protection: a defined barrier, checked only at maturity, for the legacy sleeve	~1.3 yrs	Flexibility: expected life is short, so capital returns early to reinvest at higher rates
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Own the scarce part of AI, Asia's chips, memory and tools, and get paid while the market swings.

Call and loss probabilities on the team outlook (slide 9); fair coupon from the Monte Carlo pricing (Appendix E). Not a guarantee of future returns; capital at risk below the 70% barrier and to Natixis credit.